

Technology Stack

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, Jinja2	Lightweight, responsive UI for the applicant form and dashboard, rendered directly by Flask.
2	Backend / Server-Side	Python, Flask	Integrates easily with the ML stack and offers a simple, lightweight framework for real-time predictions.
3	Database / Data Storage	MySQL	Strong relational mapping for structured applicant records and easy reporting.
4	Cloud / Hosting / Deployment	IBM Cloud	Scalable, cost-effective hosting for the Flask app and trained model.
5	Version Control & CI/CD	GitHub, GitHub Actions	Enables team collaboration and automates testing/deployment on code updates.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy	Handle data preprocessing, model training/evaluation, and prediction logic.